

Research

Understanding the adoption of renewable energy technologies by households in South Asia: a theory of planned behavior perspective

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Abstract

This study examines the factors influencing the adoption of renewable energy technologies (RETs) by households in South Asia, using the Theory of Planned Behavior (TPB) framework. Data were collected from 520 respondents across Bangladesh, India, Pakistan, Sri Lanka, Nepal, Bhutan, and the Maldives. The study explores how attitudes, subjective norms, perceived behavioral control, and barriers to adoption shape individuals' intentions to adopt RETs. Results show that attitudes towards renewable energy ($\beta = 0.092$, $p < 0.05$), subjective norms ($\beta = 0.125$, $p < 0.05$), and perceived behavioral control ($\beta = 0.124$, $p < 0.05$) significantly influence adoption intentions, while barriers to adoption ($\beta = 0.546$, $p < 0.000$) negatively impact these intentions. The study highlights that while social pressures and individual perceptions of control are crucial, overcoming financial and infrastructural barriers is key to increasing RET adoption in the region. To accelerate the adoption of RETs, the study suggests financial incentives such as subsidies, low-interest loans, and tax break, particularly targeted at low-income households. Additionally, infrastructure development to improve grid connectivity and local maintenance support is necessary. The role of community-driven awareness campaigns and leveraging social norms is also emphasized to foster broader acceptance. These findings offer actionable insights for policymakers to create more effective interventions that can facilitate a transition to renewable energy in South Asia, ensuring both environmental sustainability and energy security.

Keywords Renewable energy technologies · Adoption intentions · Theory of Planned Behavior · South Asia

1 Introduction

The adoption of renewable energy technologies (RETs) by households plays a pivotal role in advancing sustainable development and combating climate change, especially in regions such as South Asia, where energy demands are rapidly increasing. South Asia, home to over 1.7 billion people, faces significant challenges in ensuring energy security while mitigating the environmental impacts associated with conventional energy sources. Households, as the primary energy consumers, are essential to reducing reliance on fossil fuels and embracing sustainable alternatives such as solar panels, wind turbines, and biogas systems [14]. However, despite the increasing global focus on renewable energy, the adoption of RETs in South Asia remains slow and uneven, hindered by various socio-economic, infrastructural, and cultural barriers [10].

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The importance of RET adoption cannot be overstated, as it offers a pathway to reduce greenhouse gas emissions, enhance energy security, and promote sustainable livelihoods. Yet, households in the region face several challenges that hinder the widespread adoption of RETs. These challenges include high upfront costs, limited access to financial incentives, lack of awareness, inadequate infrastructure, and resistance to new technologies [22, 38, 39]. Additionally, the absence of supportive policies and financial frameworks, coupled with technical and maintenance challenges, creates further obstacles for households seeking to transition to renewable energy [4, 14]. Despite these barriers, the potential benefits of RET adoption—ranging from cost savings and enhanced energy security to reduced environmental impact—underscore the need for effective policy interventions and behaviorally informed strategies.

This study seeks to explore the factors influencing the adoption of RETs in South Asia, applying the Theory of Planned Behavior (TPB) as the conceptual framework. The TPB, developed by Ajzen [3], suggests that individuals' behavioral intentions are shaped by three primary factors: attitudes toward the behavior, subjective norms, and perceived behavioral control. Although TPB has been widely used to study pro-environmental behaviors such as energy conservation and the adoption of green technologies, its application to RET adoption in South Asia remains under-explored. Existing research on renewable energy adoption has predominantly focused on economic and technological factors, with limited attention given to the psychological and social influences that shape household decisions in the context of South Asia [8, 61]. This study addresses this gap by examining how attitudes, subjective norms, and perceived behavioral control interact to influence household intentions to adopt RETs in South Asia.

The novelty of this research lies in its application of TPB across a diverse set of countries in South Asia, including Bangladesh, India, Pakistan, Sri Lanka, Nepal, Bhutan, and the Maldives. These countries present varied socio-economic, cultural, and infrastructural contexts, making them ideal for a comparative analysis. For example, while India has made significant strides in promoting solar energy through government schemes such as the Solar Rooftop Scheme, countries like Bangladesh still struggle with affordability and access despite ongoing efforts to promote solar home systems [22]. Understanding the factors that drive or hinder RET adoption in such diverse settings is crucial for developing region-specific policies that encourage wider uptake.

This study makes a significant contribution to the literature by providing a comprehensive analysis of the factors influencing RET adoption intentions in South Asia, combining both psychological and structural elements. It extends the application of TPB to a context that has been less explored, offering new insights into the interplay between attitudes, social norms, and perceived control in shaping energy-related behaviors. Furthermore, by highlighting the importance of overcoming barriers to adoption—such as financial constraints and technical challenges—this research offers practical policy recommendations for accelerating the transition to renewable energy in the region.

In summary, this study addresses the critical research gap in understanding the behavioral drivers of RET adoption in South Asia. It contributes to the growing body of literature on sustainable energy transitions by applying a well-established theoretical framework to a region facing unique energy challenges. The findings are expected to inform policymaking, providing actionable insights for designing targeted interventions that promotes renewable energy adoption at the household level.

2 Literature review

The adoption of renewable energy technologies (RETs) by households in South Asia is a growing area of interest due to its potential to address energy security, environmental concerns, and economic challenges. However, despite the increasing recognition of RETs as a sustainable energy solution, the uptake remains slow, influenced by a combination of economic, social, and infrastructural factors [10, 22]. A large body of research has focused on the economic determinants of RET adoption, such as subsidies, financial incentives, and the affordability of technologies [38, 39, 61]. These studies highlight the significant role of financial barriers, but there is less attention given to the social and psychological factors that shape individual adoption decisions, particularly in South Asia's unique cultural context.

The Theory of Planned Behavior (TPB) has been widely applied in environmental studies, including energy conservation and green technology adoption [3, 8]. TPB posits that behavior is primarily influenced by three constructs: attitudes toward the behavior, subjective norms, and perceived behavioral control (PBC). In the context of RET adoption, attitudes reflect the individual's evaluation of renewable energy, subjective norms refer to the social pressures that encourage or discourage adoption, and PBC encompasses the perceived ease or difficulty of adopting RETs [3]. However, while these factors have been explored in different regions, there is a limited application of TPB in the South Asian context, where social and cultural dynamics play a significant role in shaping individual behaviors.

Several studies have examined the role of attitudes toward RETs in adoption decisions. For example, research in India and Pakistan has shown that positive attitudes toward environmental sustainability are linked to higher adoption rates of solar energy systems [4, 47]. Yet, studies focusing on South Asia have largely neglected the specific social factors that influence these attitudes. In particular, subjective norms, or the influence of family, community, and peer networks, have been underexplored in the region's adoption patterns [29, 65]. In collectivist societies like those in South Asia, societal approval can strongly impact an individual's decision to adopt green technologies, yet the role of social influence remains insufficiently addressed in the literature [19, 33].

Perceived behavioral control (PBC) has also been shown to significantly affect the adoption of renewable energy technologies. PBC reflects an individual's belief in their ability to overcome barriers such as high costs, technical challenges, and lack of access to RETs [48, 64]. Research in South Asia has identified that technical and financial barriers are particularly salient, but the psychological aspect of control—how individuals perceive their ability to overcome these barriers—remains less examined [32]. This gap is critical, as understanding how PBC interacts with the other two constructs of TPB can offer insights into how to support households in overcoming adoption barriers.

Despite these insights, the literature reveals several gaps. First, there is a lack of empirical studies examining the interaction between attitudes, subjective norms, and PBC specifically within South Asian households. While each of these factors has been explored individually, their combined effect on adoption intentions in this context remains under-researched. Second, the influence of subjective norms has often been treated as secondary to attitudes and PBC, but in South Asia, social influence may be just as important in shaping adoption behavior [29]. Third, although barriers to adoption have been identified, few studies have explored how these barriers impact perceived behavioral control, which could inform more effective interventions.

While TPB provides a useful framework for understanding renewable energy adoption, this study offers a meaningful extension of the theory by exploring its application in the unique socio-cultural context of South Asia. This research contributes to TPB by demonstrating that subjective norms and perceived behavioral control play a more significant role in the adoption of RETs than previously recognized in the literature. By incorporating contextual variables, such as social norms and technical barriers, this study enhances TPB's explanatory power in South Asia, where social and infrastructural factors are key to shaping adoption decisions.

The findings of this study also refine TPB by highlighting the relative importance of barriers to adoption. The significant negative impact of barriers on adoption intentions suggests that TPB could be extended to better account for the structural and economic factors that shape behavioral control in developing regions. This theoretical enhancement provides a more nuanced understanding of how psychological and contextual factors interact to influence the adoption of renewable energy technologies.

Therefore, while the literature has extensively explored the economic and technological aspects of renewable energy adoption, psychological and social factors, particularly subjective norms and perceived behavioral control, remain under-explored in the South Asian context. This study addresses these gaps by applying TPB in a region-specific context and offering a comprehensive examination of the factors that shape household intentions to adopt RETs. By focusing on the interaction between these factors and emphasizing the importance of overcoming barriers, this research offers valuable insights for policy interventions aimed at accelerating the adoption of renewable energy technologies in South Asia.

2.1 Research framework

The adoption of renewable energy technologies (RETs) by households is a complex process influenced by psychological, social, and contextual factors. The Theory of Planned Behavior (TPB) serves as the theoretical foundation for this study, offering a structured approach to understanding household decision-making. TPB posits that behavior is primarily driven by three constructs: attitudes toward the behavior, subjective norms, and perceived behavioral control [3, 7]. Each construct contributes uniquely to shaping behavioral intentions and subsequent actions, making TPB a suitable framework for exploring the adoption of RETs in South Asia. The proposed framework applies TPB to analyze the adoption of RETs, integrating contextual variables specific to South Asia, such as economic incentives, cultural norms, and policy support in Fig. 1. The subsequent literature review delves deeper into these phenomena, providing detailed insights and explanations.

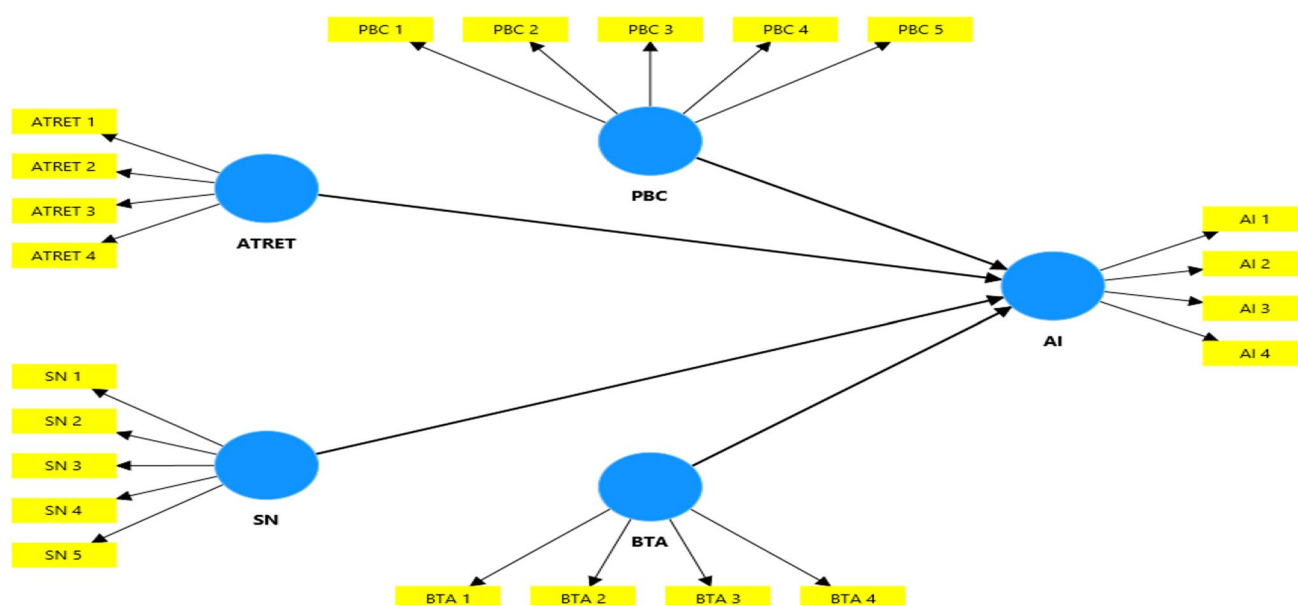


Fig. 1 Research model for the study

2.2 Justification of the model

The Theory of Planned Behavior (TPB) is highly relevant in this research design as it provides a robust framework to understand the psychological and social factors influencing individuals' intentions to adopt renewable energy technologies (RETs). TPB's constructs—attitudes, subjective norms, and perceived behavioral control—are particularly well-suited to explore the multifaceted nature of RET adoption in South Asia, where both individual beliefs and social influences play a critical role. Alternative models, such as econometric techniques, focus more on quantitative relationships and may overlook the psychological and cultural factors that TPB addresses. Given the study's focus on behavior and decision-making processes, TPB is more appropriate as it allows for a nuanced understanding of the intentions behind RET adoption, making it a better fit for this context.

2.3 Attitudes toward renewable energy technologies

Attitudes toward renewable energy technologies (RETs) significantly influence household adoption decisions. Positive attitudes arise from the perceived benefits of RETs, such as cost savings, energy security, and environmental sustainability. Studies highlight that households with favorable perceptions of RETs are more likely to exhibit stronger adoption intentions [69]. Furthermore, environmental awareness has been identified as a critical factor shaping attitudes, as individuals who recognize the ecological benefits of RETs often prioritize them over conventional energy sources [18].

Hypothesis (H1) Attitudes toward renewable energy technologies positively influence households' intention to adopt renewable energy technologies.

2.4 Subjective norms

Subjective norms refer to the perceived social pressure to engage or not engage in a particular behavior, such as adopting renewable energy technologies (RETs). In collectivist societies like those in South Asia, social influences, including family, peers, and community leaders, play a pivotal role in shaping household decisions [29]. Positive societal approval or recommendations from trusted sources often encourage households to adopt RETs. Research demonstrates that individuals are more likely to conform to pro-environmental behaviors when they perceive that influential groups or

communities approve of these actions [69]. Moreover, cultural norms in South Asia, which emphasize collective well-being, further reinforce the importance of subjective norms in decision-making.

Hypothesis (H2) Subjective norms positively influence households' intention to adopt renewable energy technologies.

2.5 Perceived behavioral control

Perceived behavioral control (PBC) refers to an individual's perception of their ability to perform a particular behavior, influenced by available resources, knowledge, and external factors. In the context of renewable energy technologies (RETs), PBC plays a crucial role in adoption decisions. Households with sufficient financial resources, technical expertise, and access to RETs are more likely to feel capable of adopting these technologies [2].

Hypothesis (H3) Perceived behavioral control positively influences households' intention to adopt renewable energy technologies.

2.6 Barriers to adoption

Despite the increasing awareness of renewable energy technologies (RETs), several barriers hinder their adoption by households. High upfront costs, lack of financing options, and insufficient infrastructure are among the most significant obstacles. These financial and logistical challenges discourage many households, particularly in developing regions like South Asia, where disposable income and technological accessibility are limited. In addition, a lack of technical knowledge and inadequate policy support further exacerbate the problem. Households may perceive RETs as complex or unreliable without proper education and government initiatives to mitigate risks [14, 17]. Addressing these barriers through subsidies, improved infrastructure, and public awareness campaigns is crucial for accelerating adoption.

Hypothesis (H4) Barriers negatively influence households' intention to adopt renewable energy technologies.

2.7 Adoption intentions

Adoption intentions represent the motivational factors influencing households' decisions to adopt renewable energy technologies (RETs). According to the Theory of Planned Behavior (TPB), intentions are shaped by attitudes, subjective norms, and perceived behavioral control, acting as a precursor to actual behavior [3, 49]. Households with stronger intentions, driven by positive attitudes, social encouragement, and a sense of control, are more likely to adopt RETs. Empirical studies highlight that adoption intentions are bolstered by awareness of environmental benefits, perceived economic savings, and societal acceptance [69]. However, weak intentions often result from perceived risks, insufficient policy support, or lack of trust in technology. Strengthening adoption intentions requires targeted interventions such as awareness campaigns and financial incentives.

3 Methodology

3.1 Measurement

The study employs established measurement scales to ensure the reliability and validity of constructs related to renewable energy technology adoption. A seven-point Likert scale (1 = "strongly disagree" to 7 = "strongly agree") was used to capture respondents' agreement levels across constructs. This scale was chosen for its sensitivity and ability to capture nuanced responses, allowing for a more granular understanding of respondents' attitudes and intentions. Adoption Intentions (AI) was measured using four items adapted from Ajzen [3] and Venkatesh et al. [64] to assess participants' intentions to adopt renewable energy technologies. Attitudes Toward Renewable Energy Technologies (ATRET), measured through four items, reflected positive or negative evaluations of such technologies, drawing from Ajzen's [3] Theory of Planned Behavior. Subjective Norms (SN) were captured via five items to evaluate perceived social pressure to adopt, as outlined by Han et al. [26]. Perceived Behavioral Control (PBC) was assessed through five items to measure perceived ease or difficulty in adoption, aligning with Ajzen's [3] conceptualization and validated by Islam et al. [34]. Barriers to

Adoption (BTA) included four items reflecting financial, technological, and infrastructural challenges based on Irfan et al., [32] and Islam et al. [36]. To ensure robust construct validity, confirmatory factor analysis (CFA) was performed. Internal consistency was confirmed with Cronbach's alpha and composite reliability exceeding the 0.7 threshold [34, 50, 51]. Discriminant validity was established using the Fornell-Larcker criterion and HTMT ratio [28]. These measurement techniques adhere to quantitative research best practices, ensuring a comprehensive understanding of factors influencing renewable energy adoption in South Asia.

3.2 Sample characteristics

Table 1 provides an overview of the demographic characteristics of the respondents involved in the study, encompassing variables such as age, gender, education level, household income, and country of residence. The age distribution reveals that the majority of respondents fall within the 31–40 years category (26.15%), followed closely by those aged 21–30 years (24.23%) and 41–50 years (22.31%). Respondents below 20 years and those aged 51 years or more constitute smaller proportions, at 20.58% and 6.73%, respectively. This indicates a predominantly middle-aged respondent base. Gender-wise, the majority of respondents are male (67.31%), while females account for 32.69%, highlighting a gender imbalance in the sample, which may reflect the study's focus area or regional participation trends. Regarding education, most respondents have a secondary school certificate (SSC) or lower (26.92%), followed by higher secondary education (HSC) at 25.77%. Bachelor's (Honors) and Master's degree holders represent 21.73% and 20%, respectively, while Ph.D. holders are the least represented group (5.58%), suggesting a diverse educational background with a focus on

Table 1 Characteristics of respondents Source: Author's own creation

Variables	Categories	Frequency	Percentage
Age	Below 20 years	107	20.58
	21–30 years	126	24.23
	31–40 years	136	26.15
	41–50 years	116	22.31
	51 years or more	35	6.73
	Total	520	100
Gender	Male	350	67.31
	Female	170	32.69
	Total	520	100
Level of education	SSC or lower	140	26.92
	HSC	134	25.77
	Honors	113	21.73
	Master's	104	20.00
	Ph.D	29	5.58
	Total	520	100
Household income	Below 200 USD	106	20.38
	200–350 USD	125	24.04
	351–450 USD	164	31.54
	451–550 USD	88	16.92
	Above 550 USD	37	7.12
	Total	520	100
Country of residence	Bangladesh	102	19.62
	India	82	15.77
	Pakistan	78	15.00
	Sri Lanka	86	16.54
	Nepal	74	14.23
	Bhutan	54	10.38
	Maldives	44	8.46
	Total	520	100

Bold values are the total number of sample size and total percentage

foundational education levels. The household income distribution shows a concentration in the 351–450 USD category (31.54%), followed by 200–350 USD (24.04%) and below 200 USD (20.38%). Respondents with incomes above 550 USD constitute a small minority (7.12%), reflecting moderate economic profiles. Lastly, the country of residence reveals that the majority of respondents are from Bangladesh (19.62%), followed by Sri Lanka (16.54%) and India (15.77%). Smaller proportions are from Pakistan, Nepal, Bhutan, and the Maldives, indicating a geographically diverse sample but with a stronger representation from specific South Asian countries. These characteristics provide a comprehensive context for interpreting the study's findings, ensuring their relevance across diverse demographic groups.

3.3 Sample size and representation

The determination of an appropriate sample size is a cornerstone of methodological rigor, ensuring the statistical reliability and generalizability of research findings. This study utilized a sample of 520 respondents, selected using stratified random sampling to ensure representation across diverse demographic and socioeconomic segments. Stratified random sampling was employed, yet the resulting sample demonstrates uneven representation across South Asian nations. Bangladesh, Sri Lanka, and India are overrepresented, while Bhutan and the Maldives are underrepresented. This disparity likely arises from several factors. Variations in the availability and accuracy of sampling frame data across countries present a significant challenge. Furthermore, data collection difficulties, including logistical constraints, lower internet penetration, and cultural factors influencing survey participation, likely contributed to the underrepresentation of certain nations. Finally, differential response rates across countries may have exacerbated the imbalance, even if initial sampling was proportionally stratified. The sample size aligns with the guidelines provided by Krejcie and Morgan [43], which suggest a sufficient sample for large populations, and meets the requirements for robust statistical analyses, particularly in Partial Least Squares Structural Equation Modeling (PLS-SEM). Sarstedt et al. [57] emphasize that PLS-SEM necessitates a minimum of 10 respondents per measurement indicator, ensuring reliable path coefficients and model validation. Furthermore, the sample size adheres to Amato et al. [6] formula for large population sampling, ensuring a high degree of precision and confidence in the findings. It supports the complexity of the study's theoretical model, providing adequate statistical power as recommended by Islam et al. [34]. The inclusion of diverse variables such as age, gender, education, income, and regional distribution enhances the heterogeneity of the sample, improving external validity [21]. This methodological approach ensures the research achieves a high standard of quality, credibility, and applicability to real-world contexts.

3.4 Data collection procedure

The data collection procedure for this study was conducted systematically to ensure accuracy and reliability. A structured survey instrument was designed, comprising validated measurement scales tailored to the study's theoretical framework. The survey was pre-tested on a small subset of respondents to assess clarity and reliability, as suggested by Dillman et al. [16]. Following the pre-test, the finalized survey was administered through both online and offline channels to reach a diverse pool of respondents, aligning with the recommendations of Saunders, [58] for mixed-mode data collection. The data collection process spanned 4 months, ensuring adequate time to capture responses across various demographic and geographic segments. Ethical considerations, including informed consent and respondent confidentiality, were strictly adhered to [11]. The survey yielded 520 valid responses, representing diverse socioeconomic and demographic backgrounds, which is critical for enhancing the study's generalizability and robustness [23].

3.5 Data analysis

The data analysis in this study followed a rigorous and systematic approach to ensure the validity and reliability of findings. Initially, the dataset was checked for missing values and outliers using descriptive statistics and diagnostic tools [52, 53, 59]. Structural Equation Modeling (SEM) with Partial Least Squares (PLS) was employed as the primary analytical technique, leveraging its capability to handle complex models with latent constructs and small-to-moderate sample sizes [28]. The measurement model was assessed for reliability and validity through Cronbach's alpha, Composite Reliability (CR), and Average Variance Extracted (AVE), ensuring internal consistency and construct validity [21]. Discriminant validity was evaluated using the Fornell-Larcker criterion and the Heterotrait-Monotrait (HTMT) ratio. Variance Inflation Factors (VIF) were examined to ensure the absence of multicollinearity among predictors [59]. For hypothesis testing, path coefficients, standard errors, and significance levels were computed using bootstrapping procedures with 5000

resamples [12]. The model's goodness-of-fit was assessed through metrics such as the Standardized Root Mean Square Residual (SRMR) and Normed Fit Index (NFI). Hypothesis 1, stating that Attitudes toward renewable energy technologies positively influence households' intention to adopt renewable energy technologies, was tested by examining the path coefficient between ATRET and AI. The significance of the path coefficient was determined using the bootstrapped standard errors and p-values. This process was repeated for each hypothesis. These comprehensive analyses provided insights into the relationships among constructs, ensuring robust and interpretable results.

3.6 Methodological justification

This study employs Partial Least Squares Structural Equation Modeling (PLS-SEM). PLS-SEM was chosen for several key reasons. First, the primary focus of this research is on predicting adoption intentions and explaining the variance in the dependent variables related to renewable energy technology adoption, rather than achieving a specific level of model fit. PLS-SEM is particularly well-suited for such predictive modeling [24]. Second, the model under investigation is relatively complex, involving 5 latent variables (Adoption Intentions, Attitudes Toward Renewable Energy Technologies, Subjective Norms, Perceived Behavioral Control, and Barriers to Adoption) and multiple indicators for each. PLS-SEM is known to perform effectively with such complex models, even with moderate sample sizes, which is advantageous in the context of South Asian research [13]. Finally, while data transformations were employed, some variables exhibited minor deviations from perfect normality. PLS-SEM's relative robustness to non-normality provides an additional justification for its selection [27].

While Covariance-Based SEM (CB-SEM) is another widely used approach for analyzing latent variable models, it was not chosen for this study because its primary emphasis is on assessing overall model fit, which is less central to the present research objectives. Furthermore, CB-SEM's stricter requirements regarding distributional assumptions made PLS-SEM a more appropriate choice given the data characteristics.

4 Analysis and findings

4.1 Indicator reliability

Table 2 presents the outer loadings, T-values, and P-values for various constructs in the context of renewable energy adoption. All indicators demonstrate strong relationships with their respective constructs, with outer loadings above the critical threshold of 0.7, which is considered adequate for validity [37]. The Adoption Intentions (AI) indicators show loadings ranging from 0.761 to 0.893, all with exceptionally high T-values (ranging from 25.582 to 70.052) and P-values of 0.000, indicating strong and statistically significant relationships. Similarly, the Attitudes Toward Renewable Energy Technologies (ATRET) indicators, with loadings between 0.771 and 0.823, also exhibit high T-values (24.772 to 36.099) and significant P-values, underscoring the importance of these indicators in shaping attitudes [25]. For Barriers to Adoption (BTA), the outer loadings range from 0.777 to 0.851, with T-values from 24.011 to 51.583, indicating that these indicators are highly relevant to the construct. The Perceived Behavioral Control (PBC) indicators, with loadings between 0.738 and 0.845, also show consistently high T-values and P-values, reflecting the strength of these constructs [3]. Lastly, Subjective Norms (SN) indicators exhibit slightly lower loadings, with the lowest being 0.708, but still maintain statistical significance, further confirming the robustness of the model [34].

4.2 Construct reliability and validity

Table 3 provides insights into the reliability and validity of the constructs used in the study. The table reports Cronbach's alpha, Composite reliability (ρ_a), Composite reliability (ρ_c), and Average Variance Extracted (AVE), which are essential for assessing the quality of the measurement model. Cronbach's alpha values for all constructs exceed the commonly accepted threshold of 0.7, indicating good internal consistency [5, 36]. For example, AI has a Cronbach's alpha of 0.846, indicating strong reliability, while SN has a value of 0.808, suggesting adequate reliability as well. Composite reliability (ρ_a) and ρ_c values, which measure internal consistency and the overall reliability of the constructs, are also above the acceptable threshold of 0.7 [54, 55]. For instance, PBC has a ρ_a value of 0.850 and a ρ_c value of 0.893, further supporting the construct's reliability. Average Variance Extracted (AVE) is used to assess convergent validity, with values above 0.5 considered acceptable [35, 38, 39]. The AVE values

Table 2 Outer loading- T values, P values

Construct	Indicator	Outer Loading	T statistics	P values
Adoption intentions	AI 1	0.880	57.810	0.000
	AI 2	0.761	27.850	0.000
	AI 3	0.772	25.582	0.000
	AI 4	0.893	70.052	0.000
Attitudes toward renewable energy technologies	ATRET 1	0.771	24.772	0.000
	ATRET 2	0.823	31.907	0.000
	ATRET 3	0.785	27.691	0.000
	ATRET 4	0.798	36.099	0.000
Barriers to adoption	BTA 1	0.777	24.011	0.000
	BTA 2	0.797	29.750	0.000
	BTA 3	0.840	41.840	0.000
	BTA 4	0.851	51.583	0.000
Perceived behavioral control	PBC 1	0.738	23.589	0.000
	PBC 2	0.781	28.189	0.000
	PBC 3	0.826	42.604	0.000
	PBC 4	0.845	46.820	0.000
	PBC 5	0.764	28.187	0.000
Subjective norms	SN 1	0.752	25.301	0.000
	SN 2	0.708	18.534	0.000
	SN 3	0.771	26.877	0.000
	SN 4	0.773	28.093	0.000
	SN 5	0.754	29.670	0.000

Table 3 Construct reliability and validity

	Cronbach's alpha	Composite reliability (rho_a)	Composite reliability (rho_c)	Average variance extracted (AVE)
AI	0.846	0.850	0.897	0.687
ATRET	0.805	0.807	0.872	0.631
BTA	0.833	0.837	0.889	0.667
PBC	0.850	0.852	0.893	0.627
SN	0.808	0.811	0.867	0.565

Table 4 Discriminant validity- Fornell Larcker criterion

	AI	ATRET	BTA	PBC	SN
AI	0.829				
ATRET	0.654	0.794			
BTA	0.801	0.696	0.817		
PBC	0.729	0.722	0.783	0.792	
SN	0.709	0.736	0.743	0.886	0.752

range from 0.565 for SN to 0.687 for AI, all of which meet the required threshold. AI has the highest AVE (0.687), demonstrating strong convergent validity, while SN shows the lowest AVE (0.565), indicating that improvement in the measurement model for SN may be needed. Overall, the reported values indicate good construct reliability and validity, supporting the robustness of the measurement model in explaining the adoption of renewable energy technologies.

4.3 Discriminant validity

Table 4 displays the results of the Fornell-Larcker Criterion used to assess the discriminant validity of the constructs in the study. The Fornell-Larcker Criterion is a widely accepted method for evaluating discriminant validity, which ensures that each construct is empirically distinct from the others [21]. To interpret discriminant validity using this criterion, the square root of the Average Variance Extracted (AVE) for each construct is compared with the correlations between that construct and other constructs. For discriminant validity to hold, the square root of the AVE for each construct should be greater than its correlations with other constructs [34]. For instance, the square root of the AVE for Adoption Intentions (AI) is 0.829 (from Table 3), and the correlations between AI and the other constructs (ATRET, BTA, PBC, and SN) are 0.654, 0.801, 0.729, and 0.709, respectively. Since 0.829 is greater than these correlation values, AI demonstrates discriminant validity. Similarly, ATRET has a square root of AVE of 0.794, which is higher than its correlations with other constructs (e.g., 0.654 with AI and 0.696 with BTA), confirming its discriminant validity. Barriers to Adoption (BTA), Perceived Behavioral Control (PBC), and Subjective Norms (SN) all meet the Fornell-Larcker Criterion, indicating that they are distinct constructs. For example, PBC has a square root of AVE of 0.792, which is higher than its correlations with other constructs, such as 0.783 with BTA and 0.886 with SN. The results from Table 4 provide strong evidence for the discriminant validity of the constructs in the model, suggesting that they represent unique dimensions in explaining renewable energy adoption.

4.4 Multicollinearity test

Table 5 presents the Variance Inflation Factor (VIF) values for the indicators of each construct in the study, which is a crucial diagnostic for identifying multicollinearity issues in structural equation modeling (SEM). Multicollinearity occurs when predictor variables are highly correlated, potentially leading to unreliable estimates and hindering the interpretability of regression coefficients [41]. The VIF measures the extent to which the variance of a regression coefficient is inflated due to collinearity with other predictors. A VIF value greater than 5 or 10 is typically considered an indicator of problematic multicollinearity [50, 51]. For the Adoption Intentions (AI) construct, the VIF values for all indicators are relatively low, ranging from 1.584 to 2.910. These values are well below the commonly accepted threshold of 5, suggesting that there

Table 5 Collinearity statistics (VIF)

Indicators of each construct	VIF
AI 1	2.694
AI 2	1.584
AI 3	1.643
AI 4	2.910
ATRET 1	1.548
ATRET 2	1.822
ATRET 3	1.607
ATRET 4	1.597
BTA 1	1.602
BTA 2	1.730
BTA 3	2.017
BTA 4	2.090
PBC 1	1.609
PBC 2	1.825
PBC 3	2.145
PBC 4	2.188
PBC 5	1.772
SN 1	1.602
SN 2	1.509
SN 3	1.624
SN 4	1.645
SN 5	1.527

Table 6 Model fit indices for the saturated and estimated models

	Saturated model	Estimated model
SRMR	0.078	0.078
d_ULS	1.556	1.556
d_G	1.582	1.582
Chi-square	3029.184	3029.184
NFI	0.637	0.637

Table 7 Path coefficients-mean, STDEV, T values, P values

	Original sample (O)	Sample mean (M)	Standard deviation (STDEV)	T statistics (O/STDEV)	P values
ATRET → AI	0.092	0.093	0.049	1.891	0.000
SN → AI	0.125	0.128	0.072	1.740	0.000
PBC → AI	0.124	0.123	0.069	1.800	0.000
BTA → AI	0.546	0.545	0.050	10.829	0.000

is no significant multicollinearity issue among the indicators of AI [44, 54, 60]. Similarly, Attitudes Toward Renewable Energy Technologies (ATRET) indicators exhibit VIF values between 1.548 and 1.822, indicating that collinearity among them is not a concern. The highest VIF value for ATRET is 1.822, which remains well below the threshold, affirming the reliability of the measurement model for this construct. Barriers to Adoption (BTA) also shows acceptable VIF values, with the range being 1.602 to 2.090. These values suggest that the indicators of BTA are not excessively correlated with one another, ensuring the validity of the latent variable in explaining adoption behavior [25, 37]. The Perceived Behavioral Control (PBC) indicators have VIF values ranging from 1.609 to 2.188, which further supports the absence of multicollinearity issues, as these values are still below the critical level of 5 [20]. Similarly, Subjective Norms (SN) have VIF values ranging from 1.509 to 1.645, which are also within acceptable limits for SEM analysis [52, 53]. Overall, the VIF results in Table 5 indicate that the model does not suffer from problematic multicollinearity, which strengthens the confidence in the robustness and validity of the relationships between the constructs.

4.5 Model fit indices

Table 6 presents the model fit indices for both the saturated model and the estimated model, which are crucial for evaluating the quality of the structural equation model (SEM). The fit indices assessed include Standardized Root Mean Residual (SRMR), d_ULS, d_G, Chi-square, and Normed Fit Index (NFI). SRMR values of 0.078 for both the saturated and estimated models indicate a good fit, as values below 0.08 are generally considered acceptable [30]. The d_ULS and d_G values, both reported at 1.556, are indices used to assess the discrepancy between the models, with lower values indicating better fit [28, 50, 51]. The chi-square statistic is equal for both models at 3029.184, suggesting that the models fit the data equally well [9], although it is important to note that chi-square is sensitive to sample size, and thus, it is often used in conjunction with other indices like SRMR or NFI for comprehensive model evaluation. The NFI value of 0.637 indicates a moderate fit, as values above 0.90 are typically preferred [9, 52, 53]. Although this NFI value is somewhat lower than ideal, the overall model fit, based on the other indices, remains acceptable, supporting the adequacy of the estimated model in representing the data.

4.6 Hypotheses testing

Table 7 presents the path coefficients, mean values, standard deviations (STDEV), t-statistics, and p-values for the relationships between various constructs—Attitudes Toward Renewable Energy Technologies (ATRET), Subjective Norms (SN), Perceived Behavioral Control (PBC), Barriers to Adoption (BTA), and Adoption Intentions (AI). The results offer both direct and indirect influences on adoption intentions, shedding light on decision-making processes related to renewable energy adoption in South Asia. These findings both confirm and challenge previous studies, offering valuable insights for policy and practice.

The path coefficient between ATRET and AI ($\beta = 0.092$, $t = 1.891$, $p < 0.05$) reveals a positive, albeit modest, effect of individuals' attitudes toward renewable energy technologies on their intentions to adopt. This result is consistent with Ajzen's Theory of Planned Behavior (TPB), which posits that attitudes are significant predictors of behavioral intentions [3, 49]. Although the effect is small, this aligns with previous studies that emphasize the importance of fostering positive attitudes toward renewable energy technologies to enhance adoption [26, 31, 42]. However, the relatively low coefficient suggests that merely improving attitudes may not be sufficient for widespread adoption in South Asia. Therefore, policies should consider multi-faceted approaches, such as addressing contextual factors that influence adoption, including barriers and perceived control.

Subjective Norms (SN) exhibit a stronger influence on AI ($\beta = 0.125$, $t = 1.740$, $p < 0.05$), indicating that social pressure and norms about renewable energy adoption have a significant impact on adoption intentions. This finding supports the TPB, where subjective norms are critical predictors of behavioral intentions [3, 66]. This result is consistent with research by Abrahamse et al. [1] and White et al. [67], who found that social influence is a key driver of sustainable behaviors. In South Asia, where collectivism and community values often guide decision-making [56], social influence could be more pronounced than in more individualistic societies. Policymakers should, therefore, focus on fostering social networks and community-based initiatives to encourage renewable energy adoption.

The path coefficient between PBC and AI ($\beta = 0.124$, $t = 1.800$, $p < 0.05$) further supports the TPB, indicating that individuals' perceptions of their ability to control the adoption process significantly influence their intentions. This finding resonates with Venkatesh et al. [64] and Islam et al. [37], who found that perceived control over adoption barriers, such as financial constraints and technical challenges, increases the likelihood of adoption. This result underlines the need for policies that enhance individuals' sense of control, such as financial incentives, access to technology, and technical support for renewable energy adoption. The importance of perceived behavioral control in adoption processes is also evident in studies by Yan et al. [68], who found that individuals' confidence in overcoming challenges is a crucial determinant of adoption decisions.

The most significant relationship in Table 7 is between Barriers to Adoption (BTA) and AI, with a path coefficient of 0.546 and a t-statistic of 10.829 ($p < 0.000$). This result highlights that perceived barriers—such as high costs, lack of knowledge, and regulatory challenges—have a substantial negative impact on individuals' intentions to adopt renewable energy technologies. This finding is consistent with the work of Kar et al. [40] and Irfan et al. [32], who identified that addressing barriers is critical to increasing adoption rates. Barriers related to financial constraints and infrastructural challenges are particularly prominent in the South Asian context [46], and policies aimed at reducing these obstacles could significantly improve adoption intentions. In contrast to studies in developed countries, where attitudes and subjective norms may play a more significant role [45], South Asian households face additional challenges that hinder adoption, such as affordability and availability [62].

The magnitude of the path coefficient between BTA and AI underscores the importance of addressing these barriers. Interventions that reduce financial and infrastructural barriers—such as offering subsidies, improving awareness of renewable energy technologies, and simplifying adoption processes—are likely to be most effective in promoting adoption. This finding aligns with previous studies that suggest the removal of financial, informational, and regulatory barriers can significantly enhance the uptake of green technologies [7, 49].

When comparing these findings to existing literature, it becomes clear that while the factors influencing renewable energy adoption in South Asia are similar to those found in other regions, the relative importance of each factor differs. For instance, in Western countries, attitudes and subjective norms are often the most influential drivers of adoption [15, 63], while in South Asia, barriers to adoption—particularly financial and infrastructural—are more prominent. This suggests that policies in South Asia should prioritize reducing these barriers to improve adoption rates.

In conclusion, the results from Table 7 indicate that while attitudes, subjective norms, and perceived behavioral control contribute to adoption intentions, the most influential factor is the presence of barriers to adoption. The findings of this study reinforce the importance of addressing barriers such as high costs, lack of knowledge, and regulatory challenges to increase adoption intentions. Policymakers should focus on creating financial incentives, improving public awareness, and enhancing infrastructure to remove these barriers. These actions would align with the TPB framework and the broader literature on adoption behavior, ensuring that South Asia can overcome the challenges to renewable energy adoption and move toward a more sustainable energy future. This study provides a summary of the hypothesis testing outcomes based on the SEM model depicted in Fig. 2, with the detailed results presented in Table 7.

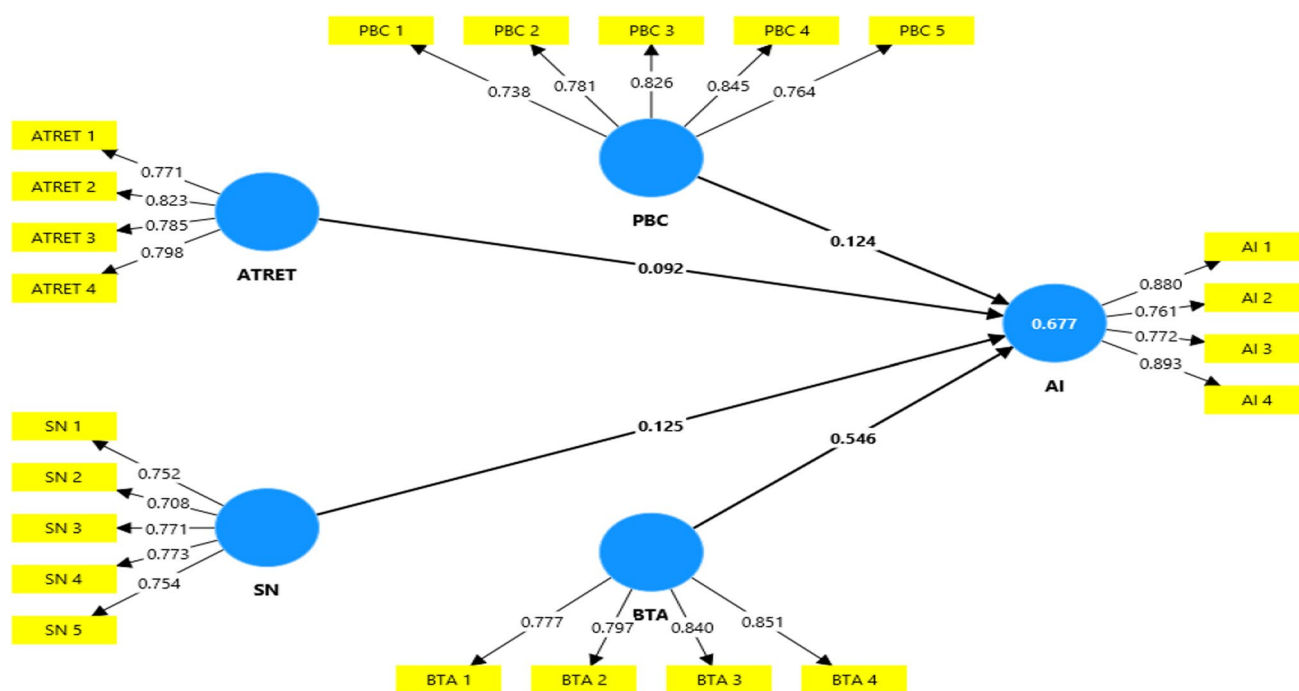


Fig. 2 The estimated structural equation model

5 Conclusion

This study offers significant contributions to understanding the factors influencing renewable energy technology (RET) adoption by households in South Asia, with a particular focus on the application of the Theory of Planned Behavior (TPB). By addressing critical gaps in the literature, this research enhances our understanding of household adoption behavior in the South Asian context, where socio-economic, cultural, and infrastructural factors shape decision-making processes. The study's findings highlight the interplay between psychological factors—such as attitudes, subjective norms, and perceived behavioral control (PBC)—and structural barriers, including financial, technological, and infrastructural challenges. These insights contribute both to the theoretical development of TPB and to the practical understanding of adoption behavior.

The theoretical contribution of this study lies in the extension of TPB within the South Asian context. While TPB has been widely used in studies of environmental behaviors, including energy conservation and the adoption of green technologies, its application to RET adoption in South Asia has been underexplored. This research refines TPB by demonstrating that subjective norms and perceived behavioral control play a more prominent role in the adoption of RETs than previously recognized. By incorporating contextual factors such as social influence and barriers to adoption, the study enhances TPB's explanatory power, offering a more comprehensive model that accounts for both psychological and structural factors. This is a significant advancement, as it challenges existing assumptions that attitudes alone are the most critical determinant of adoption in developing regions.

From a practical perspective, this study underscores the importance of multi-faceted interventions to accelerate RET adoption. The findings reveal that while positive attitudes and social influence are important, the most significant barriers to adoption—particularly financial constraints and infrastructural challenges—must be addressed to facilitate widespread uptake. Policymakers should focus on creating an enabling environment that includes financial incentives, public awareness campaigns, and the development of infrastructure to reduce perceived barriers. These practical insights are essential for guiding efforts to foster the transition to renewable energy in South Asia.

The policy implications of this study are far-reaching. The findings suggest that effective policies should prioritize overcoming the barriers to adoption, such as high initial costs and limited access to technology, through subsidies, grants, and improved infrastructure. Moreover, the role of subjective norms highlights the potential of community-based initiatives and social networks in influencing individual behaviors. Policymakers can leverage social influence

by encouraging endorsements from community leaders and creating networks that support renewable energy adoption. By focusing on both individual and structural factors, policies can be more comprehensive and effective in driving the transition to sustainable energy.

In conclusion, this study makes a significant contribution by offering a nuanced understanding of the psychological and social drivers of RET adoption in South Asia. It not only refines TPB in the region's context but also provides actionable policy recommendations that can support the widespread adoption of renewable energy technologies. The findings underscore the importance of addressing barriers while simultaneously fostering positive attitudes and leveraging social influence to achieve a sustainable energy future for South Asia.

5.1 Policy implications and suggestions

This study offers targeted policy recommendations to accelerate the adoption of renewable energy technologies (RETs) in South Asia, addressing specific barriers identified in the research. Financial incentives such as subsidies, low-interest loans, and tax breaks tailored to the region's economic conditions are essential to alleviate high initial costs, especially for low-income households, a barrier highlighted in the findings. Programs similar to India's Solar Rooftop Scheme should be expanded and adapted for diverse income groups. Additionally, enhancing grid connectivity and developing local maintenance networks are crucial to overcoming infrastructural challenges, as shown in Sri Lanka's solar integration model. Given the strong role of subjective norms in South Asian cultures, policymakers should also focus on community-based awareness programs leveraging local leaders and peer networks to promote RET adoption, supported by social media campaigns targeting younger generations. Lastly, a long-term policy framework is necessary to ensure consistent financial incentives, regulatory support, and infrastructure development, alongside capacity-building initiatives at the local level to facilitate sustainable RET use. By addressing these key areas—financial, infrastructural, social, and regulatory—policymakers can foster a supportive environment for renewable energy adoption in South Asia, overcoming barriers and ensuring a sustainable energy future for the region.

5.2 Limitations and future research

While this study provides valuable insights into the factors influencing renewable energy technology (RET) adoption in South Asia, several limitations must be acknowledged. First, the study relies on cross-sectional data, limiting the ability to draw causal inferences about the relationships between attitudes, subjective norms, perceived behavioral control, and adoption intentions. Future research could utilize longitudinal studies to track changes in adoption behavior over time and assess the long-term impact of policy interventions. Second, while the study covers a diverse range of countries in South Asia, the sample is not evenly distributed across all nations, with some countries like Bangladesh, Sri Lanka, and India being overrepresented. Future studies should aim for more balanced sampling to ensure broader generalizability of findings across the region. Additionally, this research primarily focuses on individual-level factors, with limited exploration of macro-level policy and governance issues. Future studies could delve deeper into the role of government policies, institutional frameworks, and their interaction with individual attitudes and socio-economic conditions. Finally, the study does not examine the influence of gender and intra-household dynamics on RET adoption, which could be an important area for future research, particularly in the context of South Asian households.

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Declarations

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